# Open Cluster Designs SI Status & Current Work

Loren Staley *Principal Architect, Celestica*

Lowell D. Lamb *Distinguished Engineer, Broadcom*

OCP PLATINUM® Member

# *Introduction: Open Cluster Designs for AI*

## *Who we are*

- This Strategic Initiative develops reference architectures for AI infrastructure
- Open designs covering  compute, network, storage, topology, and management
- Leverage OCP solutions and other open elements
- Incorporate contributions from leading architects across the industry

Loren Staley



## *Today's speakers*

- Loren Staley, *Principal Architect, Celestica*
- Lowell D. Lamb, *Distinguished Engineer, Broadcom*

Lowell Lamb



## *Today's agenda*

- Overview of work program & completed designs *
- Preview of designs in progress

* Published here: https://www.opencompute.org/contributions?contributions%5BrefinementList%5D%5Bproject%5D%5B0%5D=Strategic%20Initiatives%20%3E%20Open%20Cluster%20Designs%20for%20AI

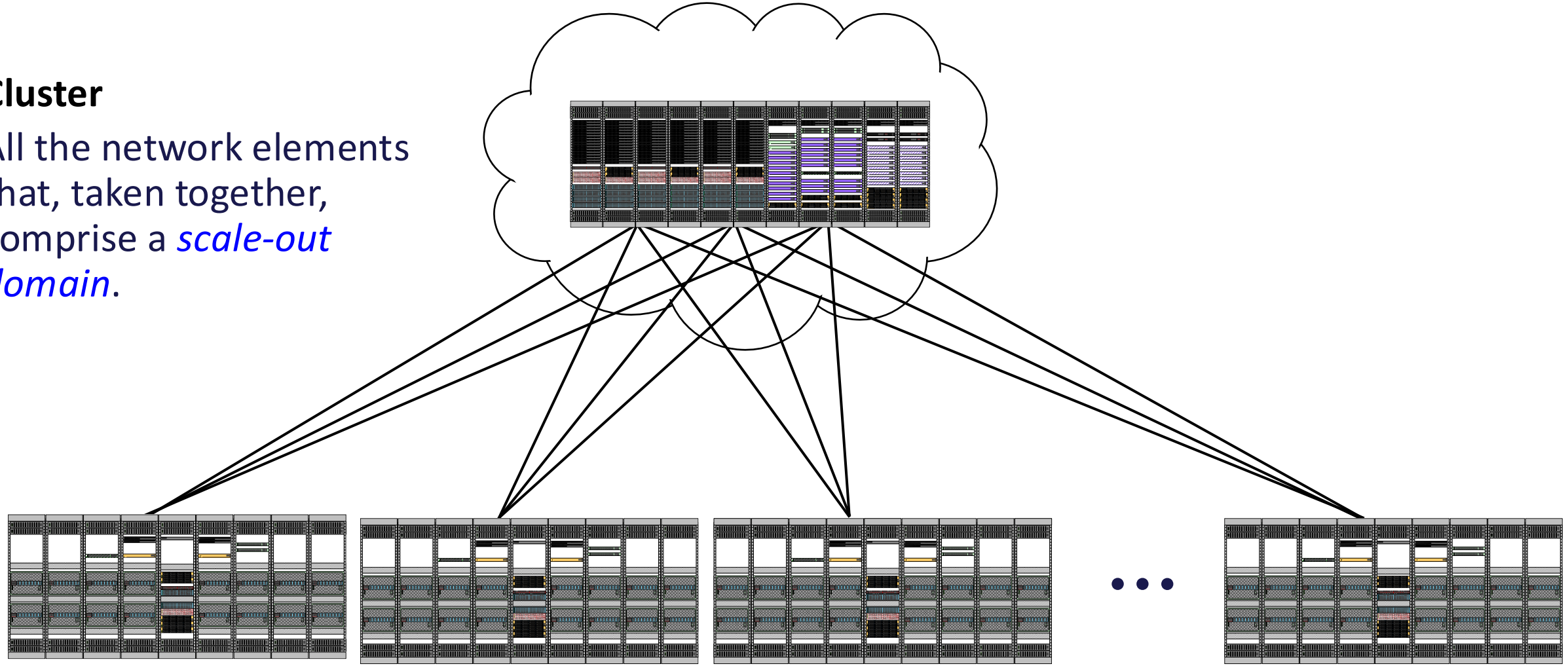
# Work program & completed designs

| **Nomenclature** | |
|---|---|
| **xPU** | Generic term for a special-purpose AI accelerator |
| **XOC-N** | xPU Open Cluster for N xPUs |
| **OPG-M** | Open Pod Group for M xPUs |

# XOC-N – *Open Cluster for N xPUs*

**Cluster**

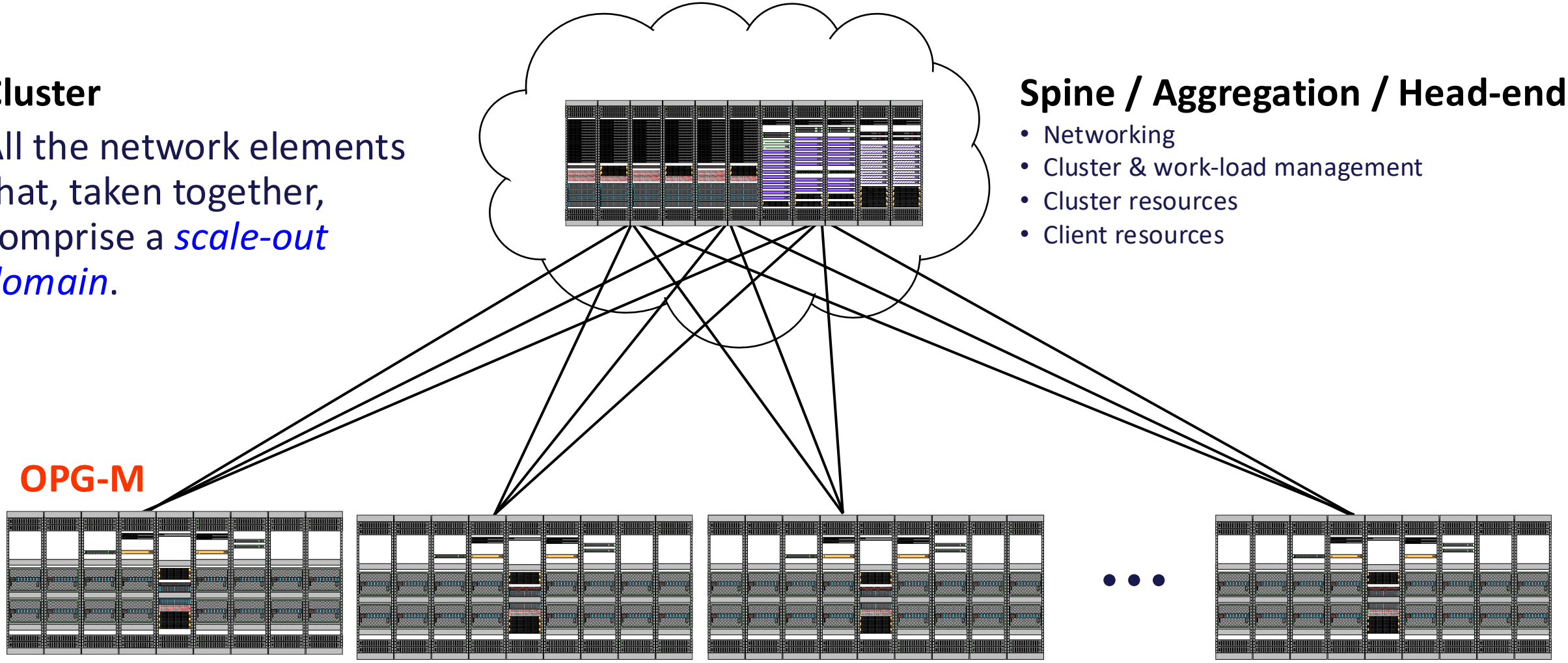
All the network elements that, taken together, comprise a *scale-out domain*.



# XOC-N – *Open Cluster for N xPUs*

**Cluster**

All the network elements that, taken together, comprise a *scale-out domain*.



**Spine / Aggregation / Head-end**

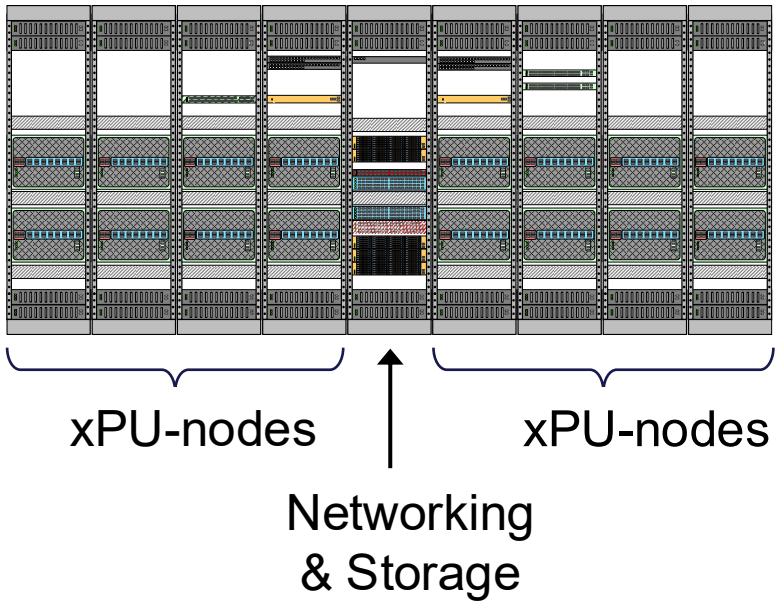
- Networking
- Cluster & work-load management
- Cluster resources
- Client resources

**OPG-M**

# OPG-M – *Open Pod Groups for M xPUs*

**400G, air-cooled OPG-128**

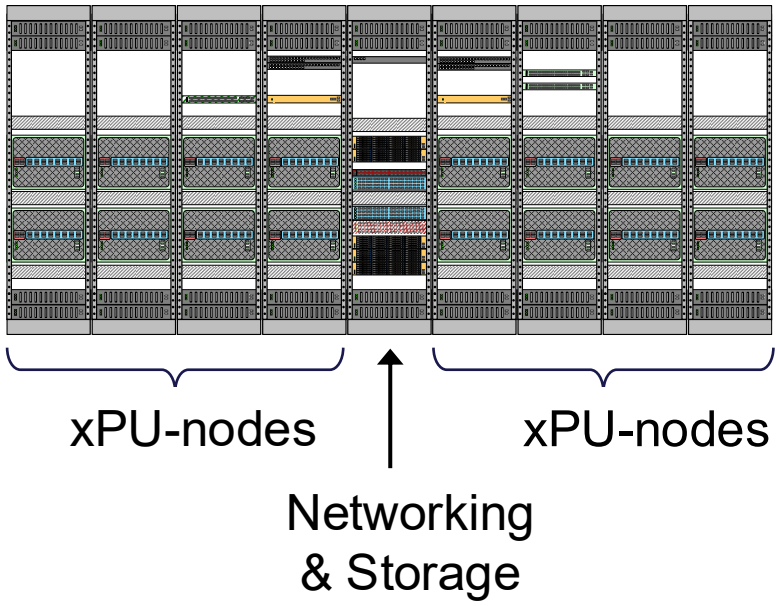
*Published January 2026*



# OPG-M – *Open Pod Groups for M xPUs*

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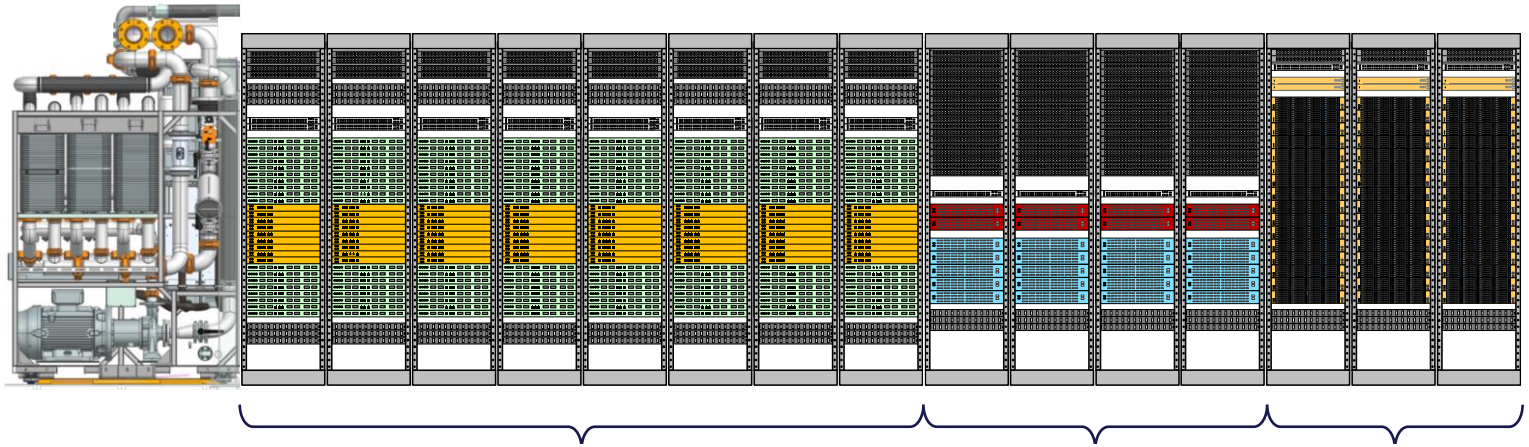
*Published January 2026*




**800G / 1.6T, liquid-cooled OPG-576 ***

*Development launched February 2026*

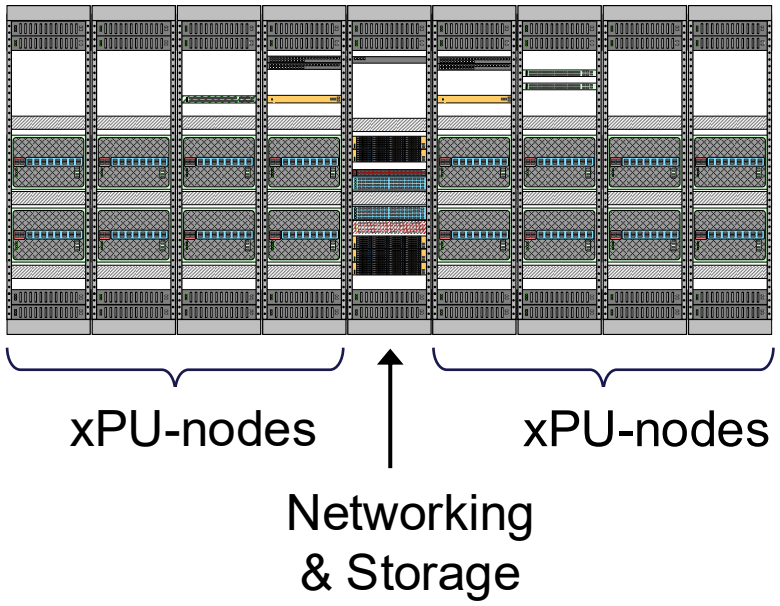
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*Published January 2026*

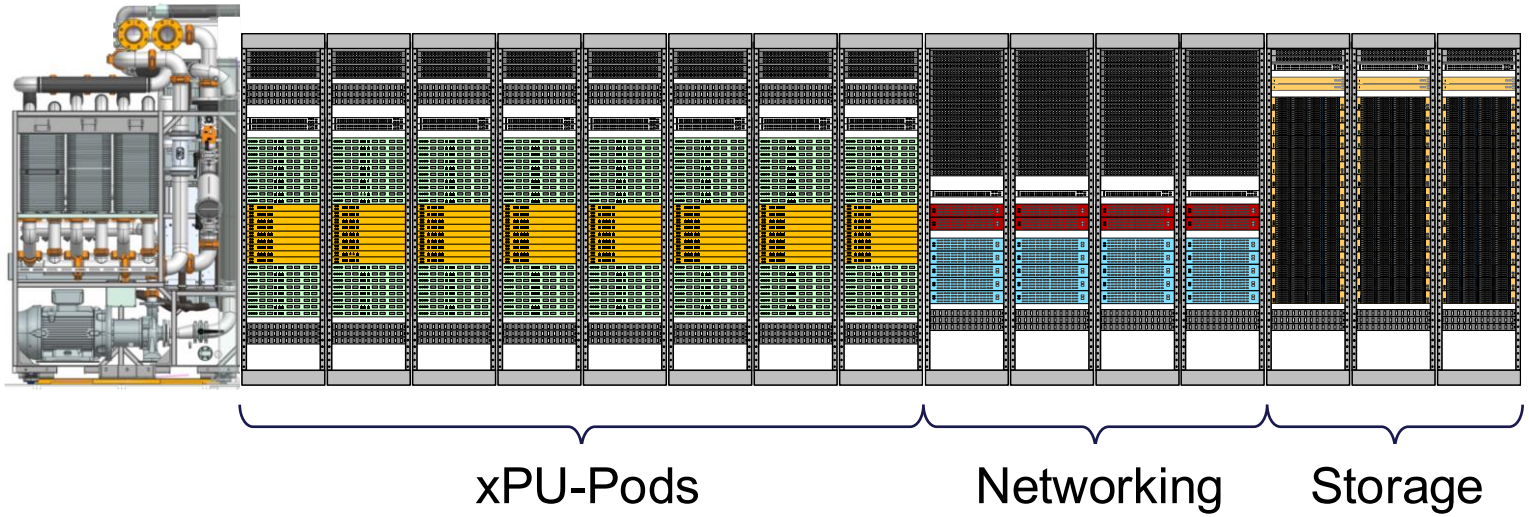



**~ 2 years**

**800G / 1.6T, liquid-cooled OPG-576 ***

*Development launched February 2026*

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Use-cases are increasingly complex and challenging

Performance targets continue to rise

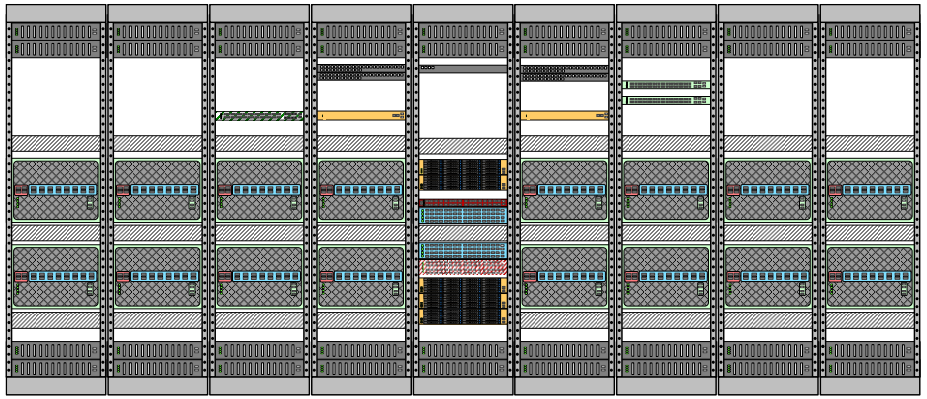
***AI infrastructure is getting bigger and more complex***

# Reference designs start with requirements

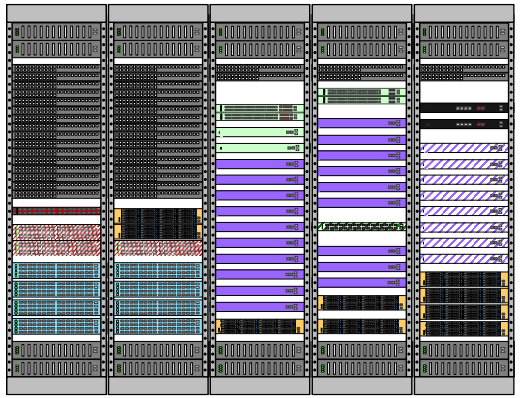
- **Use case**: training
- **Usage** (management[admin, devOps], users)
  - Clear definition of workflows and automation tasks for
    - the management: administrators and devOps
    - The end users: preprocessing, building code, building containers, submitting jobs.
- **Performance** (compute, network, storage)
  - Understanding training workloads requirements at large scale
    - Compute (CPU and xPU ratios, processing and memory)
    - Network (CPU/Network for MPI and xPU/Network for Collectives)
    - Storage (feeding datasets, checkpointing, saving model weights)
  - Balanced ratios of compute, network and storage for training workloads
- **Scalability** (compute, network, storage)
  - Designed for strong scaling: 2X,4X,.. resources → ~ 2X, ~4X, … faster runs.

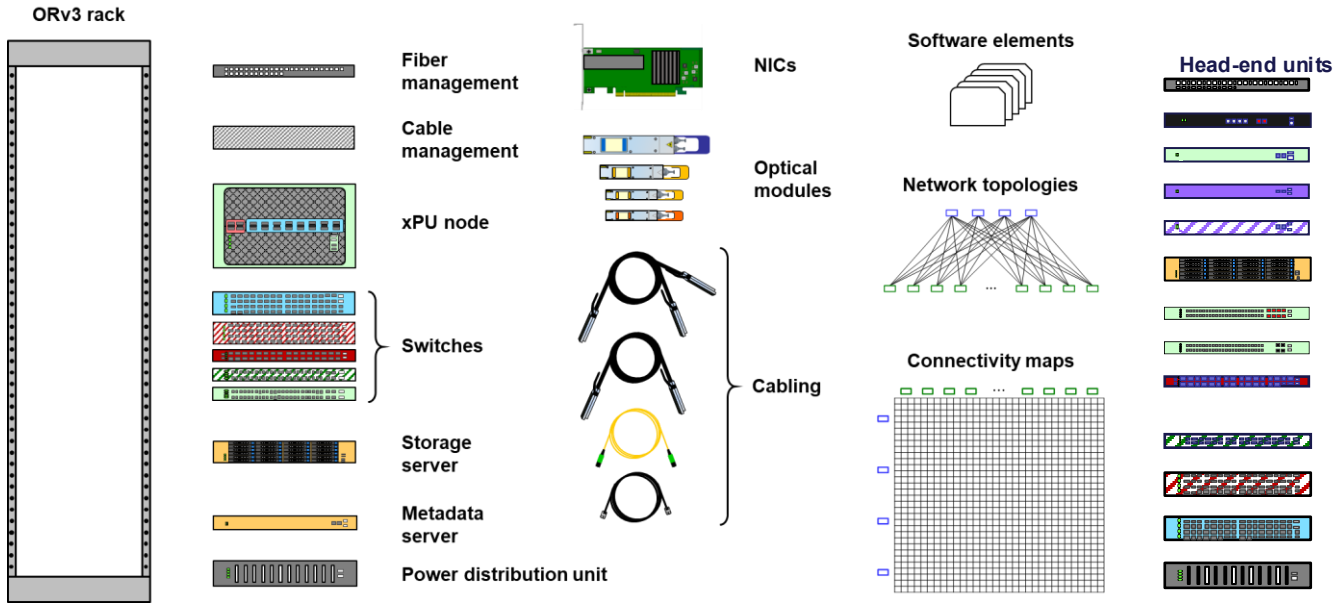
# Output: Practical specifications for AI infrastructure

## OPG (xPU racks)



## Head-end resources






## Intended for

- Design & implementation
- Procurement

### Power, space, & weight estimates

| Rack # 5 | # per rack | Unit power (kW) | Unit Height (OU) | Unit Weight (kg) | Height subtotal (OU) | Weight subtotal (kg) | Power subtotal (kW) |
|---|---|---|---|---|---|---|---|
| Storage servers | 5 | 2.0 | 2 | 38 | 10 | 190 | 10 |
| Scale-out 51 switches | 2 | 3.2 | 2 | 24 | 4 | 48 | 6.4 |
| Storage 51T switch | 1 | 3.2 | 2 | 24 | 2 | 24 | 3.2 |
| Scale-out CPU 26T switch | 1 | 1.8 | 1 | 15 | 1 | 15 | 1.8 |
| 5 kW PSU | 4 | 0.3 | 0 | 10 | 0 | 40 | 1.2 |
| Fiber management tray | 1 | 0.0 | 1 | 1 | 1 | 1 | 0 |
| Horizontal cable tray | 3 | 0 | 2 | 3 | 6 | 9 | 0 |
| DAC cables | 90.8 | | | 0.8 | | 72.6 | |
| ORv3 rack | 1 | 0 | 0 | 170 | | 170 | |
| Space total (OU) | | | | | 24 | | |
| Weigth total kg | | | | | | 570 | |
| Total maximum power (W) | | | | | | | 22.6 |
| Total de-rated power (W) | | | | | | | 18.1 |

### Hardware kits

| Item | Number |
|---|---|
| **Intrastructure** | |
| OR v3 racks | 9 |
| 5.8 kW power supplies | 36 |
| Horizontal cable management | Design choice Note 2 |
| Fiber management | Design choice Note 3 |
| | |
| **Nodes / servers** | |
| xPU nodes | 16 |
| Data-storage servers | 5 |
| Metadata servers | 2 |
| Data-storage servers | 5 |
| Metadata servers | 2 |
| | |
| **Ethernet switches** | |
| 51T Scale-out leaf (64x OSFP) | 4 |
| 26T Scale out-CPU leaf (32x OSFP) | 1 |
| 51T Storage leaf (64x OSFP) | 1 |
| 2T In-band leaf (48x 25G/SFP28) | 1 |
| Out-of-band leaf (48x 1G/RJ45) | 2 |

| Item | Number |
|---|---|
| **Optical Modules** | |
| 800G single-mode OSFP (2x 400G) | 84 |
| 100G single-mode QSFP28 | 6 |
| 10G single-mode SFP+ | 7 |
| | |
| **Optical fiber patch cables** | |
| MTP-8/12 single-mode | 168 |
| LC single-mode | 13 |
| | |
| **Optical-fiber cable break-out** | |
| Cable assembly or modular | Design choice Note 4 |
| | |
| **Passive DAC cables** | |
| OSFP <==> 2x 400G QSFP112 | 76 |
| OSFP <==> 4x 100G QSFP112 | 16 |
| SFP28 <==> SFP 28 | 23 |
| | |
| **Copper patch cables** | |
| Cat6a jumper | 64 |

https://www.opencompute.org/contributions?contributions%5BrefinementList%5D%5Bproject%5D%5B0%5D=Strategic%20Initiatives%20%3E%20Open%20Cluster%20Designs%20for%20AI

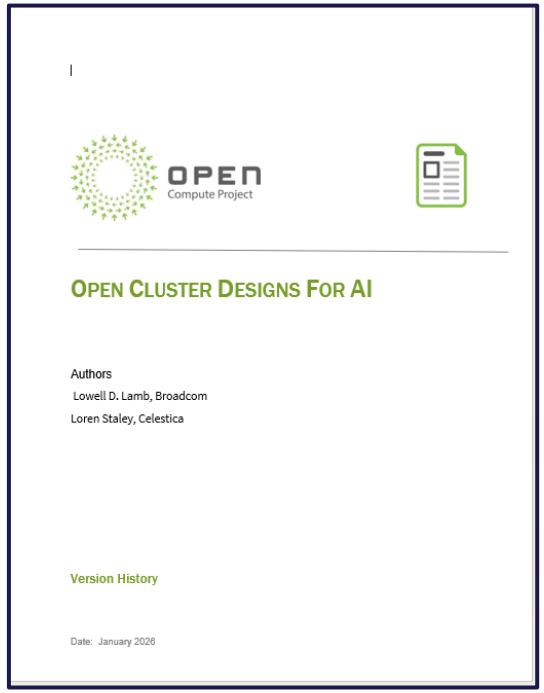
# Contributions to date

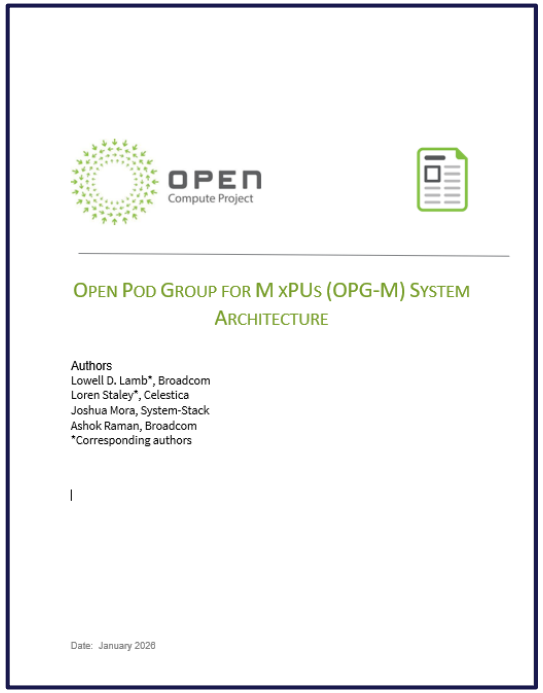



- Broadcom, Celestica, & System-Stack
- White Paper, XOC-N, & OPG-M designs
- Training use-case
- 400G SO links
- Air-cooled servers
- Publication: January 2026

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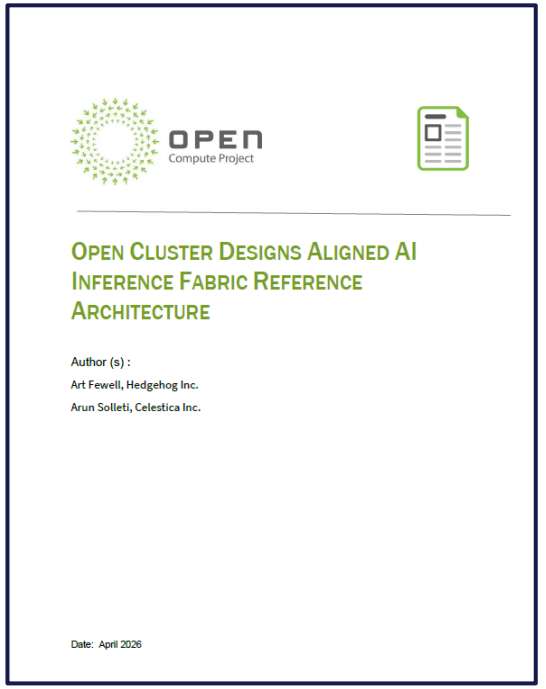
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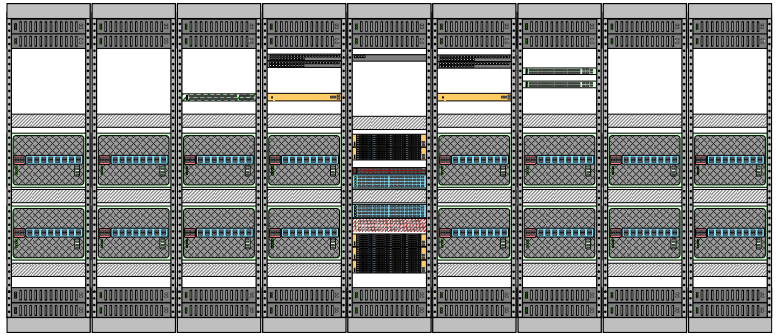
- Hedgehog & Celestica
- Open-source cluster management solution
- Target hardware: XOC-N / OPG-M platform
- Built on top of Kubernetes & SONiC
- Publication: April 2026

https://www.opencompute.org/contributions?contributions%5BrefinementList%5D%5Bproject%5D%5B0%5D=Strategic%20Initiatives%20%3E%20Open%20Cluster%20Designs%20for%20AI

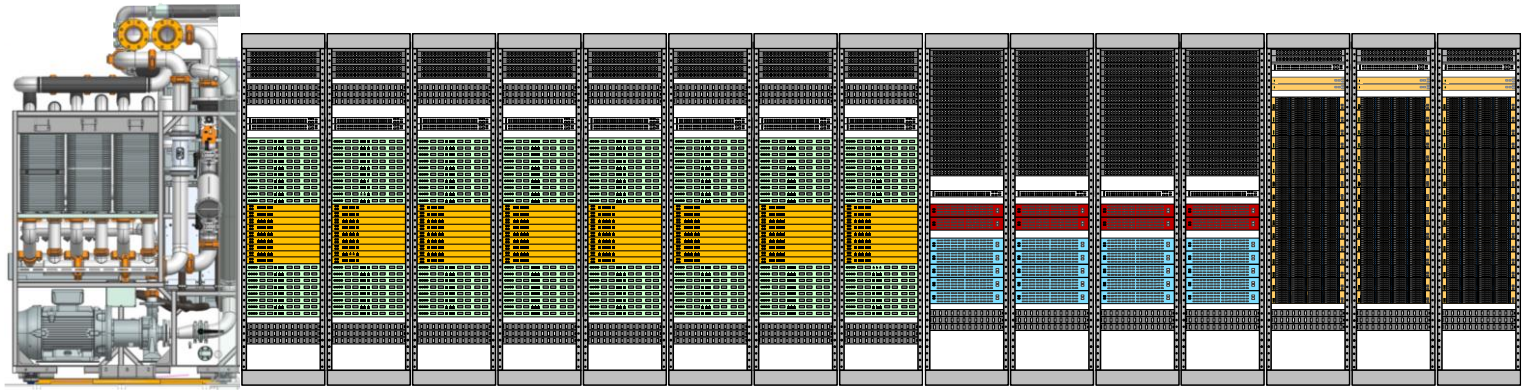
# Current work

# Problem statement

**400G scale-out links**
**air-cooled**



**800G or 1.6T scale-out links**
**liquid-cooled**



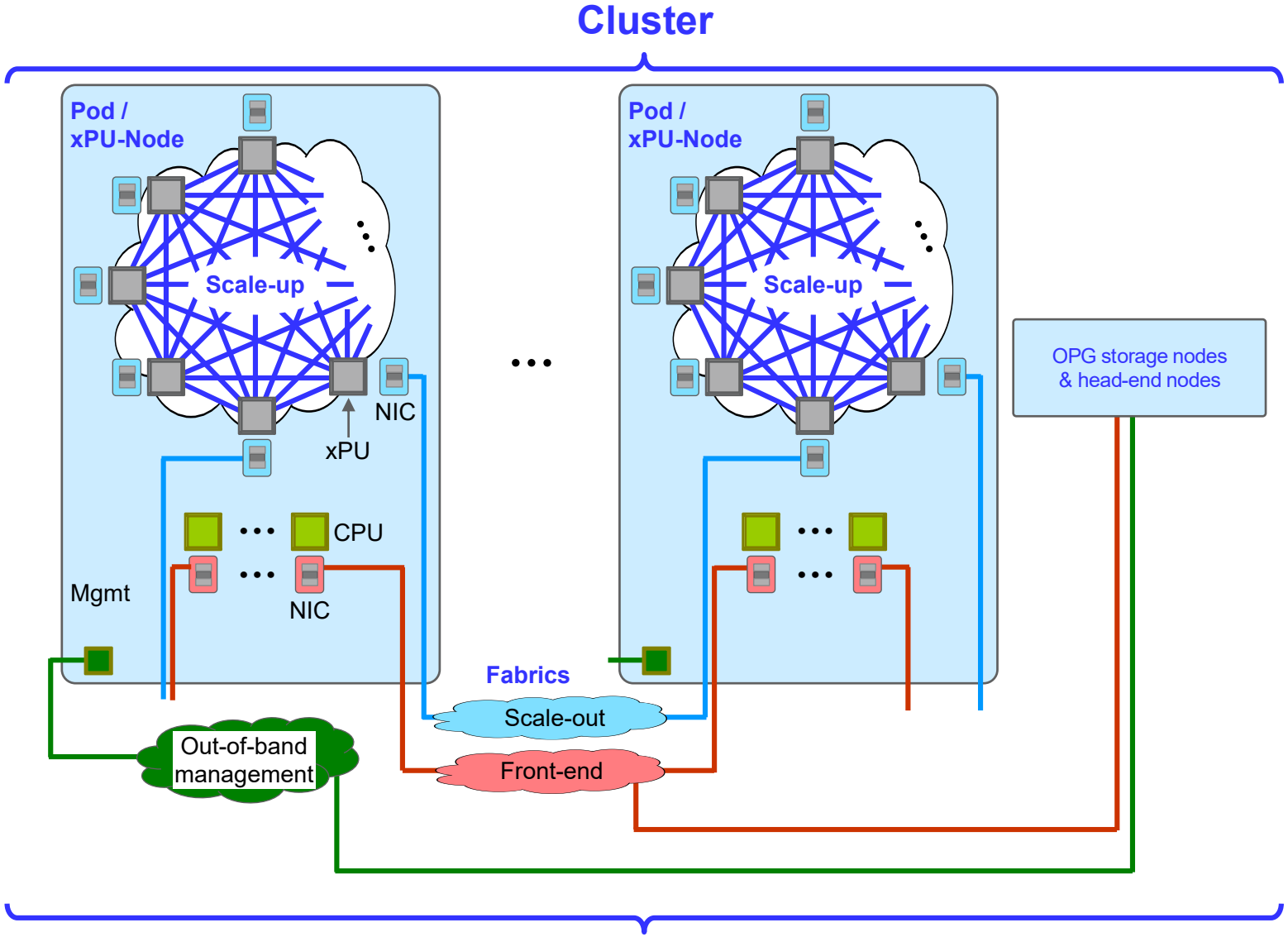
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## This is a very big transition.

What is the impact on the cluster network architecture?

What is the impact of new use cases?

# The network architecture now is simpler & faster




**Scale-out links**
800G or 1.6T per xPU

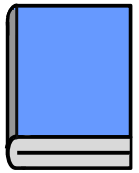
**Front-end links**
2x 400 Gb/s front-end per tray
Per xPU: ~ 100 Gb/s

*Note:* the role of the front-end network has grown in complexity.

# Document summarization example

**Prompt**

Provide a 10-page summary of *Portrait of a Lady* by Henry James.

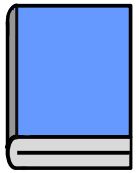


**Append a copy**
**~ 310k tokens**

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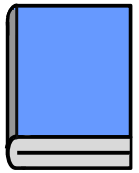
**Sentence sample**

The house had a name and a history; the old gentleman taking his tea would have been delighted to tell you these things: how it had been built under Edward the Sixth, had offered a night's hospitality to the great Elizabeth (whose august person had extended itself upon a huge, magnificent and terribly angular bed which still formed the principal honour of the sleeping apartments), had been a good deal bruised and defaced in Cromwell's wars, and then, under the Restoration, repaired and much enlarged; and how, finally, after having been remodelled and disfigured in the eighteenth century, it had passed into the careful keeping of a shrewd American banker, who had bought it originally because (owing to circumstances too complicated to set forth) it was offered at a great bargain: bought it with much grumbling at its ugliness, its antiquity, its incommodity, and who now, at the end of twenty years, had become conscious of a real aesthetic passion for it, so that he knew all its points and would tell you just where to stand to see them in combination and just the hour when the shadows of its various protuberances—which fell so softly upon the warm, weary brickwork—were of the right measure.

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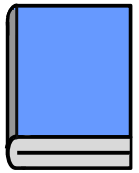
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**Question:** Ten persons are mentioned in this sentence. How are they related to each other?

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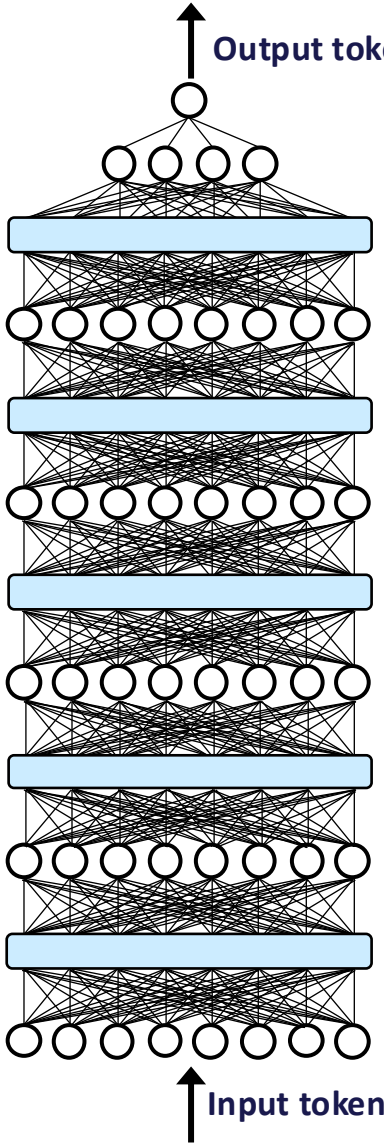
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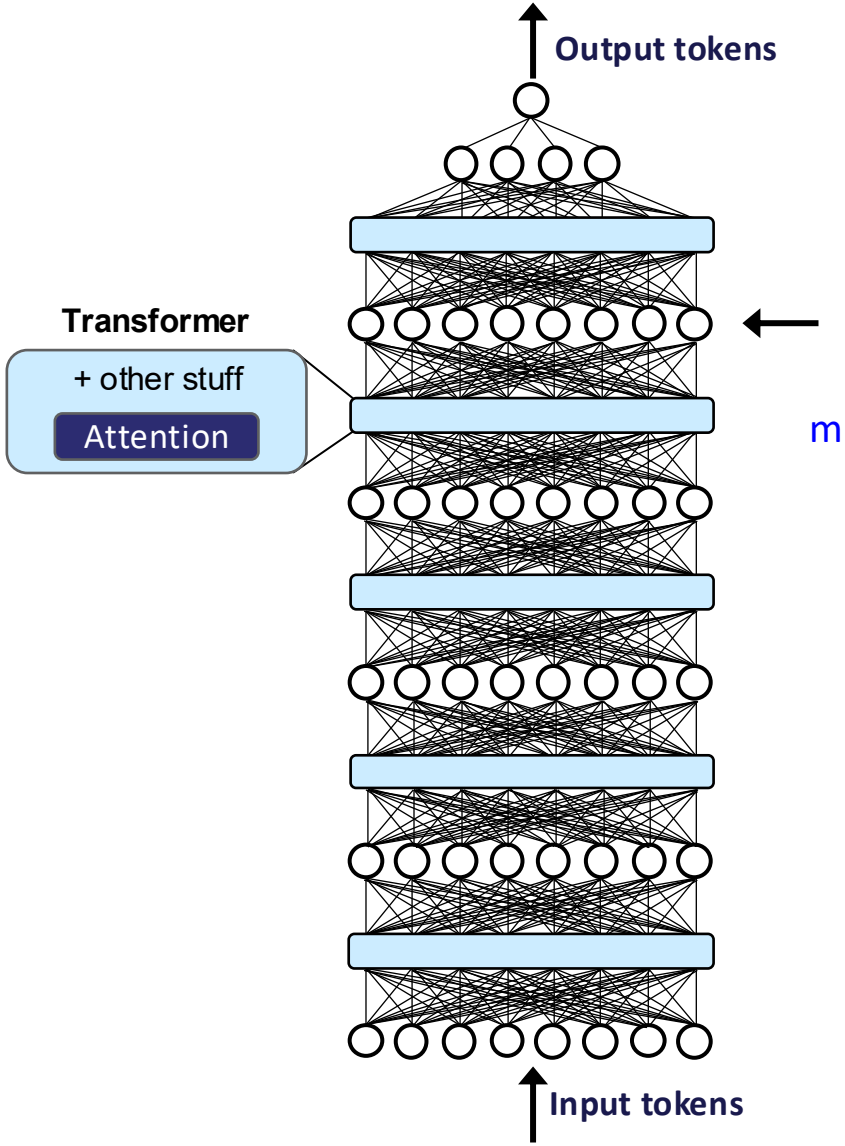
**Answer:** It depends on the ***context.*** → ***The efficient calculation and management of context information is a major challenge in AI / ML.***

# Context information is stored in the KV cache




**Feed-forward layers**
Act primarily as parametric memory storing world knowledge acquired during pre-training

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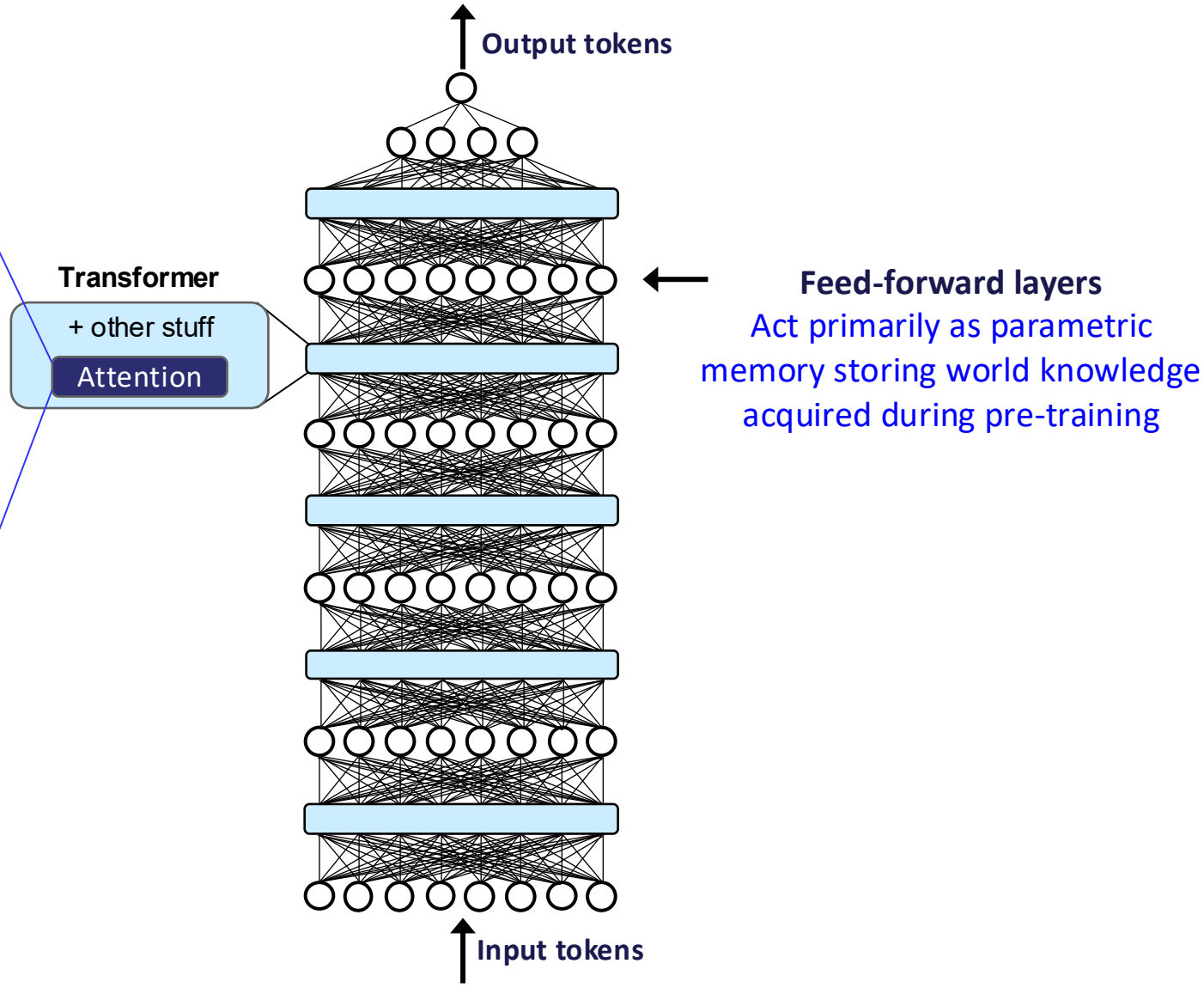


# Context information is stored in the KV cache

The Attention calculation for *each* token produces a Key (Key) and Value (V) for *each* Attention head.

These *KV pairs* are stored in the *KV cache* for use in processing all future tokens

The KV cache holds the dynamic, working *context* for the current sequence.



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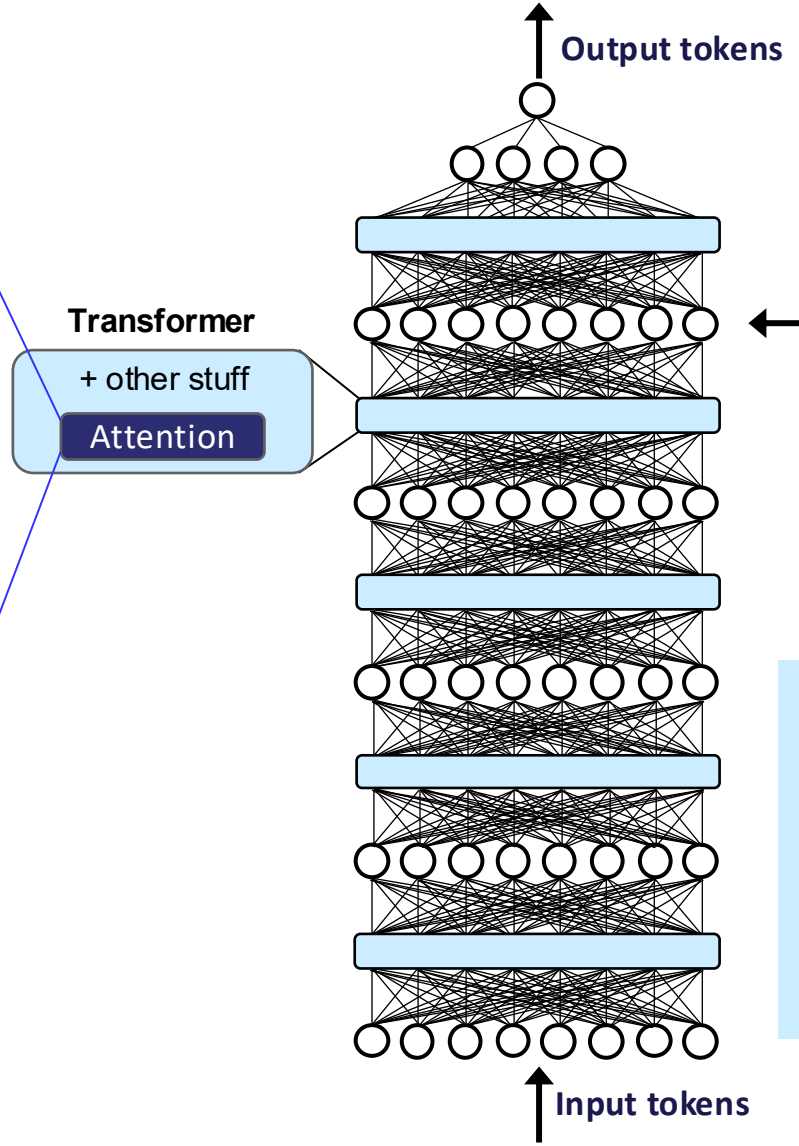
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The KV cache is

- Big (~ 10k KV pairs per token)
- Expensive to calculate
- Expensive to store
- *Indispensable*




*The efficient calculation and management of KV data is a major implementation challenge.*

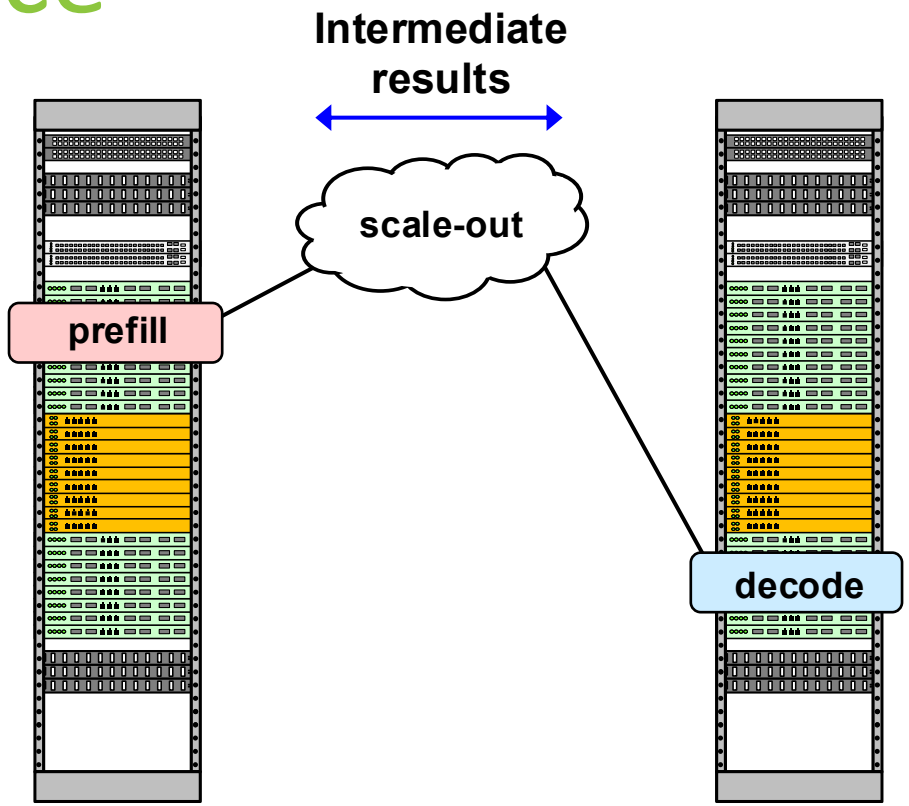
*This is one of the issues driving architectural evolution.*

# Example: disaggregated inference

**Separate the execution of an LLM into independent compute nodes**

- **Prefill:** Processes input prompts. Ingests context windows.
- **Decode:** Generates output tokens one at a time.

Processing requirements are different ➔ dissimilar HW can be used

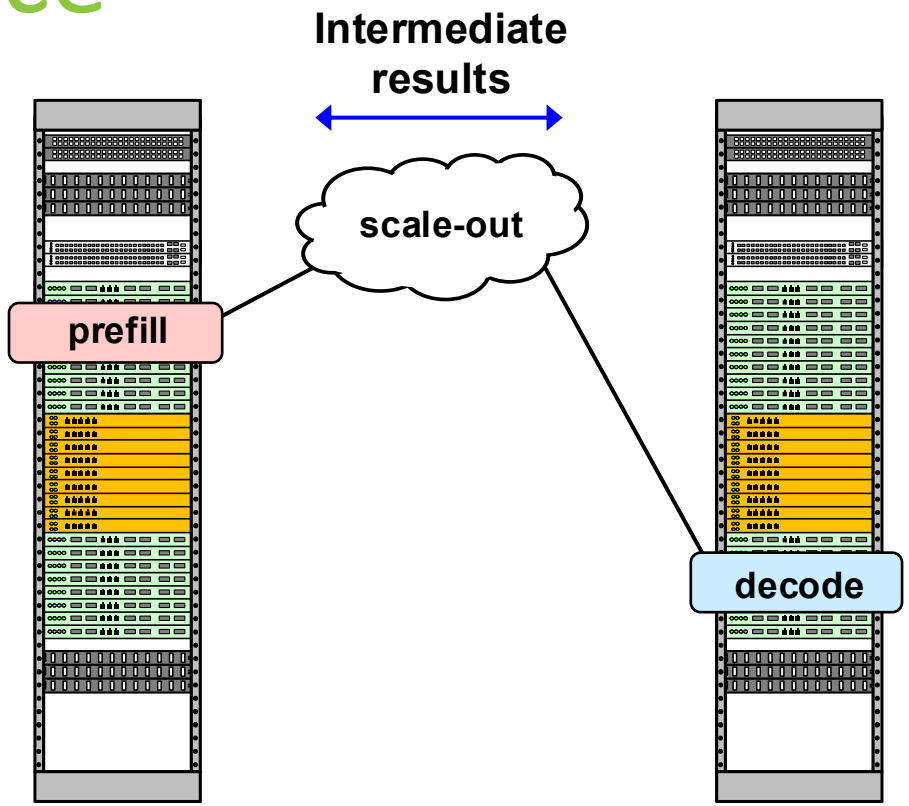


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**Different workloads lead to different prefill- / decode-instance ratios**

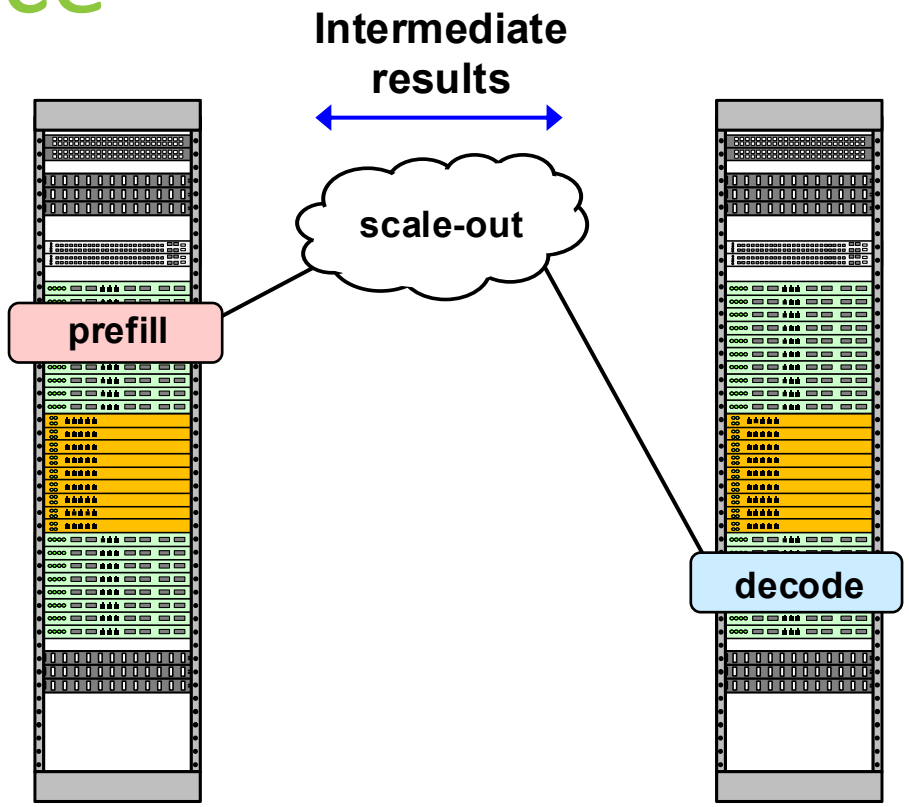
- Standard Chatbots & QA **( ~1:2 to 1:4 )**
- Reasoning Models & Agentic Chains **( ~1:3 to 1:10 )**
- High Prefix-Cache Hit Workloads **( ~1:4 to 1:8 )**
- Code Generation & Long-Form Writing **( ~1:3 to 1:6 )**
- Document Summarization & Heavy Ingestion **( ~2:1 to 4:1 )**

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**Key questions**

What are typical traffic patterns?

What are typical HW configurations?

What is a suitable SO architecture?

# Related OCP workgroups – *a sample*

- **Open Systems for AI**
  - Umbrella organization that coordinates AI work across the OCP

- **Open Data Centers for AI**
- **AI Computing Continuum**
- **Scaling AI Clusters at Neoclouds**
- **Open Platform Firmware**
- **Hardware Management**
- **Data Center Facility**
- **Rack and Power**
- **Cooling Environments**
- **Networking**
- ***Et al***

Open Compute Projects: https://www.opencompute.org/projects

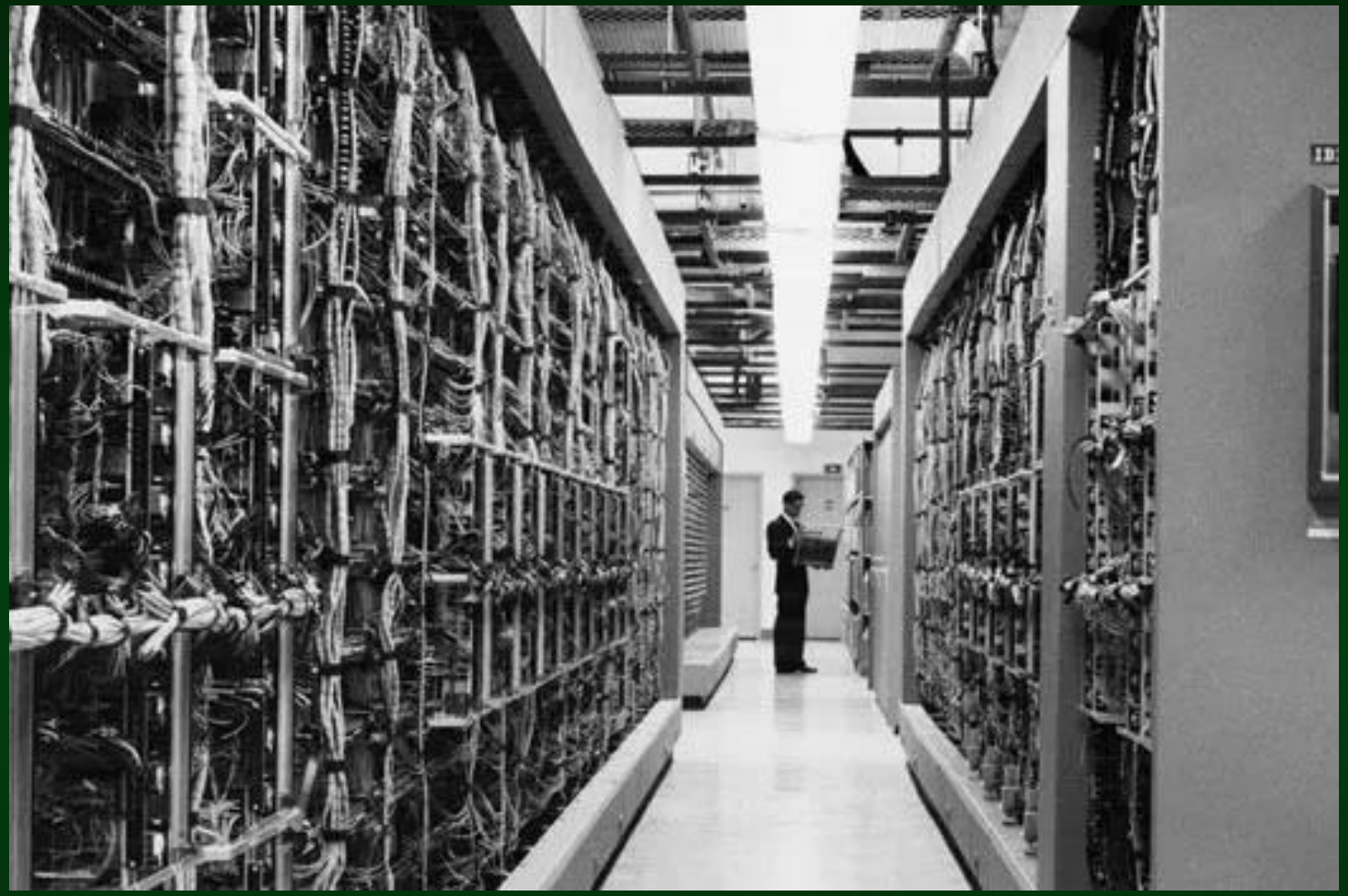
# Related OCP workgroups

- **Come join the experts** in this and other workgroups!
- **Contribute** to the growth of the AI industry

*Thank you!*

Open Compute Projects: https://www.opencompute.org/projects
*OCP Open Cluster Designs for AI* SI: https://www.opencompute.org/projects/cluster-designs-for-ai

# Thank You!



IBM's SAGE, 1956

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